

Rocketry equation

July 2026

1 rocketry equation

Nominal Operating Point

Let

$$\mathbf{x}_0 = (W_0, V_0, \Delta T_0)$$

denote the nominal launch conditions, and let

$$\varepsilon \in R, \quad |\varepsilon| \ll 1$$

represent a small perturbation from the nominal state.
Assume

$$W(\varepsilon) = W_0 \left(1 + \sum_{n=1}^{\infty} \alpha_n \varepsilon^n \right),$$

$$V(\varepsilon) = V_0 + \sum_{n=1}^{\infty} \beta_n \varepsilon^n,$$

$$\Delta T(\varepsilon) = \Delta T_0 + \sum_{n=1}^{\infty} \gamma_n \varepsilon^n.$$

The apogee model is

$$H(W, V, \Delta T) = A + B(\Delta T) + C \ln(W) + D \cos(V).$$

Substituting the perturbation expansions,

$$\begin{aligned} H(\varepsilon) = & A + B \left(\Delta T_0 + \sum_{n=1}^{\infty} \gamma_n \varepsilon^n \right) \\ & + C \ln \left[W_0 \left(1 + \sum_{n=1}^{\infty} \alpha_n \varepsilon^n \right) \right] \\ & + D \cos \left(V_0 + \sum_{n=1}^{\infty} \beta_n \varepsilon^n \right). \end{aligned}$$

Using

$$\ln(ab) = \ln(a) + \ln(b),$$

gives

$$\ln(W) = \ln(W_0) + \ln\left(1 + \sum_{n=1}^{\infty} \alpha_n \varepsilon^n\right).$$

Applying the Taylor series

$$\ln(1+x) = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{x^k}{k}, \quad |x| < 1,$$

where

$$x = \sum_{n=1}^{\infty} \alpha_n \varepsilon^n,$$

yields

$$\ln(W) = \ln(W_0) + \sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{k} \left(\sum_{n=1}^{\infty} \alpha_n \varepsilon^n \right)^k.$$

Similarly, expanding about the nominal wind speed V_0 ,

$$\begin{aligned} \cos(V) &= \cos(V_0) - \sin(V_0) \left(\sum_{n=1}^{\infty} \beta_n \varepsilon^n \right) \\ &\quad - \frac{\cos(V_0)}{2!} \left(\sum_{n=1}^{\infty} \beta_n \varepsilon^n \right)^2 \\ &\quad + \frac{\sin(V_0)}{3!} \left(\sum_{n=1}^{\infty} \beta_n \varepsilon^n \right)^3 + \dots \end{aligned}$$

Hence,

$$\begin{aligned} H(\varepsilon) &= A + B\Delta T_0 + C \ln(W_0) + D \cos(V_0) \\ &\quad + \sum_{n=1}^{\infty} K_n \varepsilon^n, \end{aligned}$$

where

$$K_n = B\gamma_n + CL_n + DM_n,$$

with

$$L_n$$

the coefficient of ε^n from the logarithmic expansion, and

$$M_n$$

the coefficient of ε^n from the cosine expansion.
Finally,

$$H(\varepsilon) = K_0 + \sum_{n=1}^{\infty} K_n \varepsilon^n$$

where

$$K_0 = A + B\Delta T_0 + C \ln(W_0) + D \cos(V_0).$$

Local Sensitivity with Respect to Weight

The apogee model is

$$H(W, V, \Delta T) = A + B(\Delta T) + C \ln(W) + D \cos(V).$$

The first partial derivative with respect to weight is

$$\frac{\partial H}{\partial W} = \frac{C}{W}.$$

Hence, the local sensitivity of the apogee to weight is inversely proportional to the rocket weight. In particular,

$$\lim_{W \rightarrow 0^+} \frac{\partial H}{\partial W} = \infty,$$

indicating that small variations in lighter rockets produce significantly larger changes in the predicted apogee.

Local Curvature

Differentiating a second time,

$$\frac{\partial^2 H}{\partial W^2} = -\frac{C}{W^2}.$$

Since

$$\frac{\partial^2 H}{\partial W^2} < 0 \quad (C > 0),$$

the logarithmic contribution is locally concave, implying diminishing returns as the rocket weight increases.

Furthermore,

$$\left| \frac{\partial^2 H}{\partial W^2} \right| = \frac{|C|}{W^2},$$

which decreases quadratically with increasing weight. Thus, the model is most strongly curved in the low-weight regime and gradually becomes locally linear as W increases.

Consequently, the first-order Taylor approximation is expected to be most accurate for heavier rockets, while higher-order terms become increasingly important for lighter rockets due to the larger local curvature.

Launch Manifold

$$M = \{(W, V, \Delta T)\}, \quad H : M \rightarrow R,$$

$$H(W, V, \Delta T) = A + B\Delta T + C \ln(W) + D \cos(V).$$

Coordinate Map

$$u = \ln(W), \quad du = \frac{dW}{W}.$$

Hence

$$H(u, V, \Delta T) = A + B\Delta T + Cu + D \cos(V).$$

Metric

$$g = \frac{dW^2}{W^2} + \frac{dV^2}{V_0^2} + \frac{d(\Delta T)^2}{T_0^2}.$$

Equivalently,

$$g = du^2 + \frac{dV^2}{V_0^2} + \frac{d(\Delta T)^2}{T_0^2}.$$

Christoffel Symbol

$$g_{WW} = \frac{1}{W^2},$$

$$\Gamma_{WW}^W = \frac{1}{2} g^{WW} (2\partial_W g_{WW} - \partial_W g_{WW}) = -\frac{1}{W}.$$

Geodesic Equation

$$\frac{d^2 x^k}{ds^2} + \Gamma_{ij}^k \frac{dx^i}{ds} \frac{dx^j}{ds} = 0.$$

For the weight coordinate,

$$\frac{d^2 W}{ds^2} - \frac{1}{W} \left(\frac{dW}{ds} \right)^2 = 0.$$

Covariant Hessian

$$\nabla_i \nabla_j H = \partial_i \partial_j H - \Gamma_{ij}^k \partial_k H.$$

$$\frac{\partial H}{\partial W} = \frac{C}{W},$$

$$\frac{\partial^2 H}{\partial W^2} = -\frac{C}{W^2},$$

$$\frac{\partial^2 H}{\partial V^2} = -D \cos(V),$$

$$\frac{\partial^2 H}{\partial (\Delta T)^2} = 0.$$

Covariant Expansion

$$H(p) = H(p_0) + \nabla_i H \xi^i + \frac{1}{2} \nabla_i \nabla_j H \xi^i \xi^j + O(\|\xi\|^3).$$

application

Apogee Sensitivity Model

$$H = H(W, V, \Delta T)$$

$$\Delta H \approx \nabla H \cdot \Delta x + \frac{1}{2} \Delta x^T (\nabla^2 H) \Delta x$$

where

$$\Delta x = \begin{bmatrix} \Delta W \\ \Delta V \\ \Delta T \end{bmatrix}.$$

Sensitivity

$$\nabla H = \begin{bmatrix} \frac{\partial H}{\partial W} \\ \frac{\partial H}{\partial V} \\ \frac{\partial H}{\partial \Delta T} \end{bmatrix}$$

$$S_i = \left| \frac{\partial H}{\partial x_i} \right|$$

$$\eta_i = \frac{S_i}{\sum_j S_j}$$

Uncertainty Propagation

$$\sigma_H^2 = \nabla H^T \Sigma \nabla H$$

where

$$\Sigma = \begin{bmatrix} \sigma_W^2 & 0 & 0 \\ 0 & \sigma_V^2 & 0 \\ 0 & 0 & \sigma_T^2 \end{bmatrix}.$$

Optimization

$$\max H(W, V, \Delta T)$$

$$\nabla H = 0$$

$$\nabla^2 H \begin{cases} < 0, & \text{maximum} \\ > 0, & \text{minimum} \end{cases}$$

Launch Condition Manifold

$$M = \{(W, V, \Delta T)\}$$

$$H : M \rightarrow R$$

$$ds^2 = \frac{dW^2}{W^2} + \frac{dV^2}{V_0^2} + \frac{d(\Delta T)^2}{T_0^2}$$

Geodesics

$$\frac{d^2 x^k}{ds^2} + \Gamma_{ij}^k \frac{dx^i}{ds} \frac{dx^j}{ds} = 0$$

Curvature

$$\nabla_i \nabla_j H = \partial_i \partial_j H - \Gamma_{ij}^k \partial_k H$$

$$\frac{\partial^2 H}{\partial W^2} = -\frac{C}{W^2}$$

$$\frac{\partial^2 H}{\partial V^2} = -D \cos(V)$$

Performance Metric

$$J(W,V,\Delta T) = \frac{H(W,V,\Delta T)}{W}$$

$$\max J$$

$$\nabla J = 0$$

Solve

1. Data Collection

Given experimental/simulation data:

$$(W_i, V_i, \Delta T_i, H_i), \quad i = 1, 2, \dots, n$$

where

$$H_i = A + B\Delta T_i + C \ln(W_i) + D \cos(V_i).$$

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2. Matrix Form

Define

$$X = \begin{bmatrix} 1 & \Delta T_1 & \ln(W_1) & \cos(V_1) \\ 1 & \Delta T_2 & \ln(W_2) & \cos(V_2) \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \Delta T_n & \ln(W_n) & \cos(V_n) \end{bmatrix}$$

$$\theta = \begin{bmatrix} A \\ B \\ C \\ D \end{bmatrix}$$

$$Y = \begin{bmatrix} H_1 \\ H_2 \\ \vdots \\ H_n \end{bmatrix}.$$

Therefore,

$$X\theta = Y.$$

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3. Least Squares Solution

For

$$n > 4,$$

solve:

$$\theta = (X^T X)^{-1} X^T Y$$

giving

$$(A, B, C, D)$$

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4. Construct Model

$$H(W, V, \Delta T) = A + B\Delta T + C \ln(W) + D \cos(V)$$

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5. Sensitivity

$$\nabla H = \begin{bmatrix} \frac{\partial H}{\partial W} \\ \frac{\partial H}{\partial V} \\ \frac{\partial H}{\partial \Delta T} \end{bmatrix}$$

$$\nabla H = \begin{bmatrix} \frac{C}{\overline{W}} \\ -D \sin(V) \\ B \end{bmatrix}$$

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6. Curvature

$$\nabla^2 H = \begin{bmatrix} -\frac{C}{\overline{W}^2} & 0 & 0 \\ 0 & -D \cos(V) & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

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7. Taylor Approximation

Around

$$x_0 = (W_0, V_0, \Delta T_0)$$

$$H(x) \approx H(x_0) + \nabla H(x_0) \Delta x + \frac{1}{2} \Delta x^T \nabla^2 H(x_0) \Delta x$$

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8. Optimization

Find

$$\nabla H = 0$$

and classify using

$$\nabla^2 H.$$

$$\text{Optimal launch condition}$$

Configuration Manifold

$$M = (0, \infty) \times R \times R$$

$$H : M \rightarrow R$$

$$H(W, V, \Delta T) = A + B \Delta T + C \ln(W) + D \cos(V)$$

Level Sets

$$\Sigma_c = H^{-1}(c) = \{x \in M : H(x) = c\}$$

$$\Sigma_c = \{(W, V, \Delta T) : A + B\Delta T + C \ln(W) + D \cos(V) = c\}$$

Regular Values

$$\nabla H \neq 0$$

$$\implies$$

Σ_c is a smooth 2-manifold.

Critical Points

$$\nabla H = 0$$

$$\begin{cases} \frac{C}{W} = 0 \\ -D \sin(V) = 0 \\ B = 0 \end{cases}$$

Hessian

$$\nabla^2 H = \begin{pmatrix} -\frac{C}{W^2} & 0 & 0 \\ 0 & -D \cos(V) & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\det(\nabla^2 H) \neq 0$$

$$\implies$$

H is Morse (locally).

Sublevel Sets

$$M_c = \{x \in M : H(x) \leq c\}$$

$$M_{c_1} \subseteq M_{c_2}, \quad c_1 < c_2$$

Boundary

$$\partial M_c = \Sigma_c = H^{-1}(c)$$

Betti Numbers

$$\beta_0 = \#(\text{connected components})$$

$$\beta_1 = \#(\text{independent loops})$$

$$\beta_2 = \#(\text{enclosed voids})$$

Critical Values

$$c^* = H(x^*), \quad \nabla H(x^*) = 0$$

$$c = c^* \implies \text{Topology Change}$$

Homotopy

$$\Sigma_{c_1} \simeq \Sigma_{c_2}, \quad c_1, c_2 \notin \{c^*\}$$

Persistent Filtration

$$= M_{c_0} \subseteq M_{c_1} \subseteq \cdots \subseteq M_{c_n} = M$$

$$\beta_k(c) = \text{rank } H_k(M_c)$$

$$\begin{array}{c} M \xrightarrow{H} R \\ \Downarrow \\ H^{-1}(c) \Longrightarrow \Sigma_c \\ \Downarrow \\ \nabla H = 0 \Longrightarrow c^* \\ \Downarrow \\ \text{Topology Transition} \end{array}$$

Time-Dependent Variables

$$W = W(t), \quad V = V(t), \quad \Delta T = \Delta T(t)$$

$$H(t) = A + B\Delta T(t) + C \ln(W(t)) + D \cos(V(t))$$

Chain Rule

$$\frac{dH}{dt} = \frac{\partial H}{\partial W} \dot{W} + \frac{\partial H}{\partial V} \dot{V} + \frac{\partial H}{\partial (\Delta T)} \dot{T}$$

$$\frac{dH}{dt} = \frac{C}{W} \dot{W} - D \sin(V) \dot{V} + B \dot{T}$$

Second Derivative

$$\frac{d^2 H}{dt^2} = -\frac{C}{W^2} \dot{W}^2 + \frac{C}{W} \ddot{W} - D \cos(V) \dot{V}^2 - D \sin(V) \ddot{V} + B \ddot{T}$$

Differential Operator

$$L = \frac{d}{dt}$$

$$LH = \frac{C}{W} \dot{W} - D \sin(V) \dot{V} + B \dot{T}$$

$$L^2H = \frac{d^2H}{dt^2}$$

Vector Field

$$X = \dot{W} \frac{\partial}{\partial W} + \dot{V} \frac{\partial}{\partial V} + \dot{T} \frac{\partial}{\partial (\Delta T)}$$

$$\mathcal{L}_X H = X(H) = \frac{dH}{dt}$$

Exponential Weight

$$\dot{W} = -kW$$

$$W(t) = W_0 e^{-kt}$$

$$\frac{C}{W} \dot{W} = -kC$$

Linear Temperature

$$\Delta T = \Delta T_0 + \alpha t$$

$$\dot{T} = \alpha$$

$$B\dot{T} = B\alpha$$

Oscillatory Wind

$$V = V_0 + \beta \sin(\omega t)$$

$$\dot{V} = \beta \omega \cos(\omega t)$$

$$-D \sin(V) \dot{V} = -D \beta \omega \sin(V) \cos(\omega t)$$

Fourier Extension

$$F(V) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nV) + b_n \sin(nV))$$

$$H = A + B\Delta T + C \ln(W) + F(V)$$

Fourier Coefficients

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} F(V) \cos(nV) dV$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} F(V) \sin(nV) dV$$

Generalized Model

$$H = A + B\Delta T + C \ln(W) + \sum_{n=1}^{\infty} (a_n \cos(nV) + b_n \sin(nV))$$

Generalized Dynamics

$$\frac{dH}{dt} = \frac{C}{W} \dot{W} + B\dot{T} + \sum_{n=1}^{\infty} n (-a_n \sin(nV) + b_n \cos(nV)) \dot{V}$$

Asymptotic Expansion

$$H = H_0 + \varepsilon H_1 + \varepsilon^2 H_2 + O(\varepsilon^3)$$

$$H \rightarrow \frac{dH}{dt} \rightarrow \frac{d^2 H}{dt^2} \rightarrow \mathcal{L}_X H \rightarrow F(V) \rightarrow \text{Fourier Dynamics}$$